Chapter 19 | Written solutions

Language Understanding

These twelve solutions follow the chapter's order: reason about representations, pooling and entity counts, verify the helpers, then design evaluations. The written PDF contains every explanation and both plots. The notebook PDF also contains the executed code and outputs.

The examples use small supplied arrays and annotations on a CPU. They do not download or train an encoder. This isolates the heads and evaluation rules from the model that would produce the features. In particular, padding invariance below holds for fixed features; it cannot establish that an encoder's attention mask is correct. Span boundaries are zero-based and half-open; tags use IOB2.

1. Find what word counts lose

Choose vocabulary order `(Ari, called, Sam)`. Both sentences have count vector $(1, 1, 1)$. A classifier receiving only that vector must receive the same evidence for both sentences, however its weights were trained.

In these simple active-voice sentences, the name before *called* is the caller and the name after it is the recipient. A Transformer with positional information can relate those positions to the verb. Ari's contextual representation can therefore reflect a different role in the two sentences.

Bidirectional attention supplies access to both sides; it does not by itself label one side as earlier. Without position information or another order-sensitive mechanism, a standard encoder's outputs permute with its inputs, and an invariant pooling operation cannot distinguish these two bags of tokens. The counts below check what was lost; they do not simulate a trained contextual encoder.

2. Pool only the valid rows

The mask selects the first three rows. The fourth contributes neither a vector to the numerator nor a row to the denominator:

$$\mathbf{h}_{\text{masked}} = \frac{(2,0)+(1,3)+(0,1)}{3} = \left(1, \frac{4}{3}\right), \qquad \mathbf{h}_{\text{all}} = \frac{(3,4)+(6,6)}{4} = \left(\frac{9}{4}, \frac{5}{2}\right).$$

Let $\mathbf{m}$ be the mean of $n > 0$ existing rows and $\mathbf{p}$ an added padding vector. Keeping the unmasked mean unchanged requires

$$\frac{n\mathbf{m}+\mathbf{p}}{n+1} = \mathbf{m} \quad \Longleftrightarrow \quad \mathbf{p} = \mathbf{m}.$$

Thus adding one row preserves the unmasked mean only when that row equals the old mean in every coordinate. For the three valid rows, that vector is $(1, 4/3)$. Zero padding generally does not satisfy the condition. Masked pooling remains unchanged for any excluded row while the valid feature rows stay fixed.

3. Assign entity tags

For *Sam left Lake City*, the word tags are `B-PER, O, B-LOC, I-LOC`. Sam starts a one-word person entity; Lake starts the location and City continues it. Use the illustrative split `La, ##ke`, not a claim about a particular tokenizer's actual vocabulary. The chapter assigns IDs 0, 1, 2, 3 to `O, B-PER, B-LOC, I-LOC`.

Token	Word ID	Supervised tag	Target ID
`[CLS]`	-1	Ignored	-100
Sam	0	B-PER	1
left	1	O	0
La	2	B-LOC	2
`##ke`	2	Ignored	-100
City	3	I-LOC	3

`O` is a real class: the model should learn that *left* is outside an entity. An ignored target supplies no direct tagging loss and is not another class. The continuation piece remains visible to attention, so its content can help the representation at `La`. At evaluation time, take predictions at Sam, left, La and City to recover exactly one tag per word.

An all-piece policy would require an explicit mapping such as `B-LOC, I-LOC` for the pieces of Lake. Copying `B-LOC` to both pieces starts two entities at subword level. Equal loss weights for all pieces also give split words more contributions than unsplit words.

4. Compare tag and span accuracy

The gold spans are $(0, 2, \mathrm{ORG})$ and $(3, 4, \mathrm{PER})$. The predicted spans are $(0, 2, \mathrm{ORG})$ and $(3, 4, \mathrm{LOC})$. The interval $[0, 2)$ includes word indices 0 and 1; its end is excluded.

Four of five tags agree, giving word accuracy $4/5 = 80\%$. Only the organization matches as a complete typed span. Thus TP = 1, FP = 1 and FN = 1. The wrong location is an unmatched prediction, and the person remains unfound. Precision, recall and F_1 are all $1/2$.

Matching only boundaries would incorrectly count both spans as correct. The single-word entity has the right location in the sentence but the wrong type.

5. Keep sentence identity when scoring

Sentence A contributes no match, one gold entity and no prediction. Sentence B contributes no match, no gold entity and one prediction. Together, $M = 0$, $P = 1$, $G = 1$. Dataset precision, recall and F_1 are all zero.

The two triples share offsets and type, but refer to different text. Include the sentence ID, as in `(A, 0, 1, PER)`, or count matches inside each sentence before adding them. Dropping the ID makes the sets intersect and falsely gives all three scores a value of one. Individual sentences have an undefined ratio when their denominator is zero; the dataset totals here have nonzero denominators.

For the separate C/D dataset, C has $F_1 = 2/(1 + 1) = 1$ and D has $F_1 = 2/(3 + 3) = 1/3$. Their mean is $2/3$. Micro F_1 first adds counts, giving $2(1 + 1)/(1 + 3 + 1 + 3) = 1/2$. The latter weights the underlying entity counts, while the former gives each sentence equal weight. Both are calculable; they answer different aggregation questions.

6. Change the padding rows

The first three rows from Exercise 2 represent tokens in a sentence. The fourth row is padding. To make a sentence prediction, the head first pools the token features, meaning it combines them into one vector. In masked mean pooling, `True` selects a row for the average and `False` leaves it out of both the sum and the count. With the Boolean mask `(True, True, True, False)`, the hand calculation is

$$\frac{(2,0)+(1,3)+(0,1)}{3}=\left(1,\frac{4}{3}\right).$$

Each logit is a raw class score. The linear head takes the dot product of the pooled vector with one row of weights per class and adds that class's bias. Here `np.eye(2)` gives the identity matrix and `np.zeros(2)` gives zero biases, so the first logit is the first pooled coordinate and the second logit is the second. Replacing the fourth row $(6,6)$ with $(20,-10)$ leaves the same three rows selected. We therefore expect the same pooled vector and logits before and after the replacement.

Marking the last row `True` makes the mask select all four rows, so the helper now computes the ordinary unmasked mean. The sum of the first three rows is $(3,4)$. With the original fourth row, the mean would be $((3+6)/4,(4+6)/4)=(9/4,5/2)$. After the change, it is

$$\frac{(3,4)+(20,-10)}{4}=\left(\frac{23}{4},-\frac{3}{2}\right).$$

Both coordinates can change because the fourth row now contributes to the sum, and the divisor is four. The identity head copies each new mean directly into the logits.

This invariance is a statement about the supplied encoder outputs. If a real encoder lets valid tokens attend to padding, changing a padded input can change the valid output rows before pooling. A pooling mask cannot repair that earlier contamination. Test the encoder attention mask separately.

7. Align a split word

There are three annotated words and seven token positions. A word ID tells us which word a token piece belongs to. The ID `-1` marks a position with no word, and consecutive pieces of the same word share its ID. In `(-1, 0, 0, 1, 2, 2, -1)`, words 0 and 2 each have two pieces. Word 1 has one piece.

With first-piece supervision, copy the label of each word to its first piece. Give the other positions the target `-100`, the default ignore value, so their own predictions contribute no direct loss. At evaluation time, read the predicted class IDs at the same first-piece positions to get one prediction per word. The table uses zero-based token positions.

Token position	Word ID	Training target	Prediction	Keep?
0	-1	-100	9	No, no word
1	0	4	4	Yes, first piece
2	0	-100	8	No, continuation
3	1	0	0	Yes, first piece
4	2	2	2	Yes, first piece
5	2	-100	7	No, continuation
6	-1	-100	9	No, no word

The three retained class IDs match their respective word labels. Reading first pieces gives a direct, word-by-word comparison with the original annotations.

8. Choose a repair policy

Read `B-LOC, I-LOC, I-PER, I-PER, O` one word at a time. In the chapter's IOB2 convention, `B-LOC` starts a location, `I-LOC` continues an open location, and `O` marks a word outside any entity. The first two tags open and extend a location. The next tag, `I-PER`, says to continue a person, but the open entity is a location. That is the first invalid continuation.

The default decoder stops at the invalid tag. Catch the exception so that the notebook records the failure and can continue to the repair call.

Word index 2 is the **third** word because indexing starts at zero. With `repair=True`, the first `I-PER` acts as a new start at index 2. The decoder closes the location at that same index and saves it before opening the person span. The next `I-PER` continues the person, and `O` at index 4 closes it.

A typed span records its start, its exclusive end, and its entity type. The half-open interval $[0, 2)$ includes word indices 0 and 1, just as the Python slice `[0:2]` does. The person covers $[2, 4)$, so the two entities are adjacent and share no words.

Set the prediction repair policy before scoring and keep gold annotations in their stated convention.

9. Test a boundary error (optional)

Use `O, B-LOC, I-LOC, O, B-PER` as the gold word tags. There are two gold entities. The location covers indices 1 and 2, and the person covers index 4. Make each prediction from a fresh copy of gold so that the three changes are independent.

Prediction	Word index	Gold tag	Predicted tag
Shorter location	2	`I-LOC`	`O`
Wrong type	4	`B-PER`	`B-ORG`
Extra entity	0	`O`	`B-ORG`

All three sequences follow IOB2. A one-word entity begins with `B`, so the organization at index 0 in the third prediction can appear immediately before the location.

Exact matching requires the start, end, and type to agree. Decode the predictions so that we can inspect the full triples. The intersection of the predicted and gold sets contains the exact matches. Let M be the number of matches, N_p the number of predictions, and N_g the number of gold entities. Precision divides by predicted spans and recall divides by gold spans.

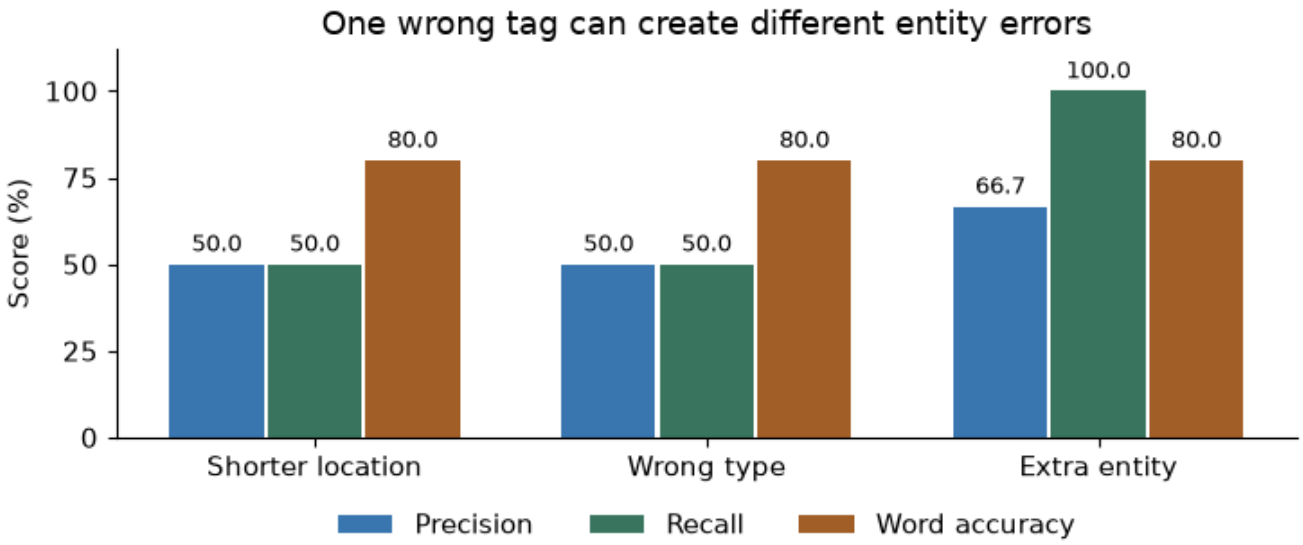
$$P = \frac{M}{N_p}, \qquad R = \frac{M}{N_g}, \qquad F_1 = \frac{2M}{N_p + N_g}.$$

Use `span_metrics` for each prediction. Check the results against the expected exact matches and the counts implied by the three tag changes.

Shorter location. The predicted location $(1, 2)$ stops before word index 2, while the gold location $(1, 3)$ includes it. The person is the only match. There is one match out of two predictions and two gold spans, so both precision and recall are $1/2$. **Wrong type.** The entity at $(4, 5)$ has the right boundaries but is predicted as `ORG`. Only the location matches. Precision and recall are again $1/2$. **Extra entity.** The organization at $(0, 1)$ adds a false positive. Both gold entities still match, so recall is $2/2 = 1$ and precision is $2/3$.

All three predictions have word accuracy $4/5$ because each changes exactly one of five tags. That single score cannot tell whether a prediction missed an entity or added one.

The plot compares the three Exercise 9 predictions. Every one changes one word tag, so all three word accuracies equal 80%. The extra entity keeps recall at 100% but lowers precision to 2/3; the boundary and type errors each lower both precision and recall to 1/2.



10. Change the label mix (optional)

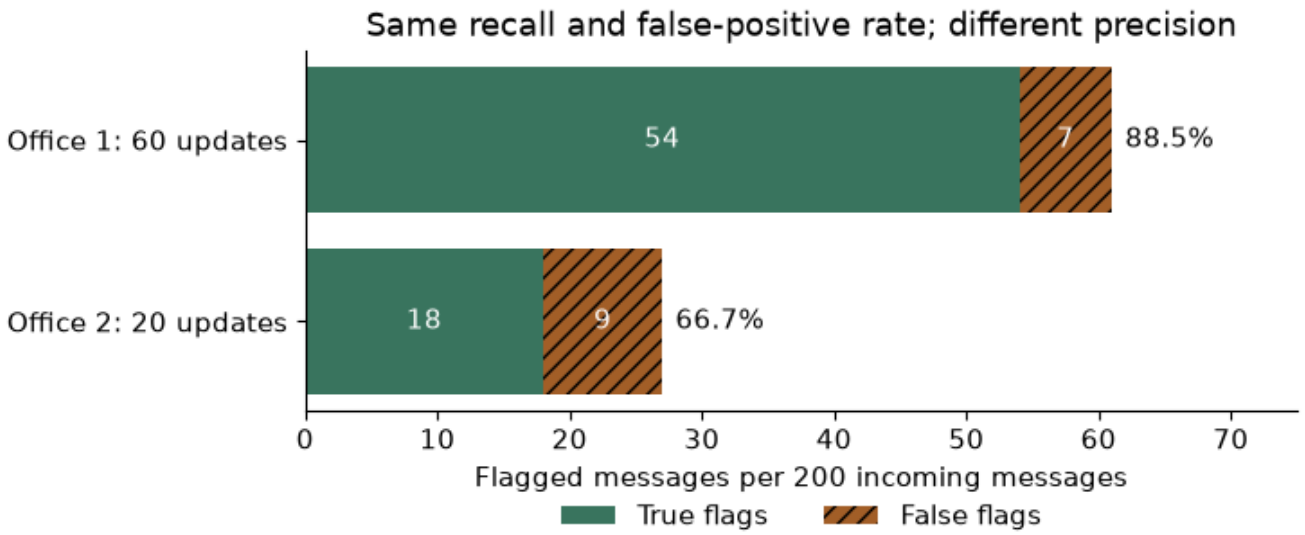
Take UPDATE as positive. Office 1 has 60 updates and 140 lookups; office 2 has 20 updates and 180 lookups. Multiply the number in each actual class by its conditional flag rate: 90% for updates and 5% for lookups.

Expected count	**Office 1**	**Office 2**
TP: updates flagged	54	18
FN: updates missed	6	2
FP: lookups flagged	7	9
TN: lookups not flagged	133	171
All flags	61	27

Precision is $54/61 \approx 88.5\%$ in office 1 and $18/27 \approx 66.7\%$ in office 2. Both have recall 90% and false-positive rate 5%. The second office has fewer actual updates to find and more lookups that can trigger false alarms. These are expected counts under the stipulated rates; finite samples can differ.

Writing π for update prevalence, r for recall and f for the false-positive rate gives population precision $r\pi/[r\pi + f(1-\pi)]$. Here π is 0.3 or 0.1. The formula follows by dividing expected true flags by expected total flags; it does not assume that conditional rates survive a real domain shift.

The figure isolates this change in prevalence. A useful additional check is recall on indirectly phrased updates, such as *I moved last week*. Apply the same annotation rule and decision threshold in both offices. Report how many such updates each contains and the fraction flagged. Aggregate recall could stay the same while performance on this important slice changes.



11. Plan an independent split (optional)

Imagine three offices using the same template, *Please send a copy of the address on file*. A customer writes to Office A and follows up with Office B. A random message split can put closely related requests on both sides. Choose the holdout unit to match the generalization claim.

To test new customers, group by stable customer ID across all offices. Assign every message and thread from a customer to one partition. Test customers must be absent from both training and validation. Keep several offices represented if the claim concerns new customers at familiar offices. Report customer and message counts, label counts and the office mix in every partition.

To test a new office, hold out that office and select models on other offices. For a test that also excludes familiar-customer shortcuts, remove from training and validation all messages from customers appearing in the held-out office. For example, if a customer used C and A, keep the C test messages and exclude their A messages from fitting and selection. Record the exclusions. This is a new-office comparison with a customer-disjoint constraint; office holdout alone does not imply new customers. Repeat separately for other held-out offices if the data supports it, and report each office rather than only a pooled result.

Audit templates as well as IDs. Compare normalized texts after replacing variable slots such as names and addresses. Then flag likely paraphrases using word or character overlap and semantic similarity, followed by manual checks. Common topic does not imply identical intent: *confirm my new address* and *enter my new address* may require different actions. Save the template-family IDs and the rule used to create them.

If the claim includes unfamiliar wording, keep each prohibited template family in one partition. Connecting records that share customer, thread or template family forms groups that enforce all those constraints together. For an intact held-out office, exclude linked customers or prohibited template families from training and validation, then check again for overlap. Report how much data this removes: strict constraints can connect much of the dataset and make the desired independent comparison unsupported. If familiar templates are allowed by the question, report that fact instead of presenting the test as unseen wording.

In the small audit, test records c1 and c2 come from Office C. Customer u1 also has record a1 in Office A, so a1 is excluded. Records a2 and b1 remain for training and validation. This checks the customer constraint; it does not establish semantic independence of the complete dataset.

12. Design an entity error study (optional)

Evaluate both taggers on the same documents, preserving the original word boundaries and gold location spans. Build the seen-name list from complete gold locations in the task training split. Normalize by a declared rule, such as case folding and collapsing spaces. A test gold name is seen if its full normalized form is in that list. This measures exposure during task training: a pretrained model may have encountered an otherwise unseen name elsewhere.

An exact match requires the same sentence, start, exclusive end and type. For diagnosis, pair unmatched predictions with unmatched gold spans in the same sentence using a fixed one-to-one overlap rule, such as greatest shared word count with a deterministic tie break. An overlapping prediction with the correct type but different boundaries is a boundary error. Identical boundaries with the wrong type are a type error. Record both differences separately as a combined error; unpaired gold spans are misses and unpaired predictions are false positives. Diagnostic pairing never converts an error into an exact match. Use the same declared IOB2 decoding or repair policy for both systems.

Assign paired errors to the slice of the complete gold name. If gold *Lake City* is unseen and the tagger predicts only *Lake*, the boundary error belongs to the unseen slice even if *Lake* alone was a training location. Assigning it by the prediction's surface form would move the error into the wrong slice. Report unpaired false positives separately because no gold name defines their slice.

For each slice, report support as the number of gold mentions, exact recall as matched gold mentions divided by support, and boundary, type, combined-error and miss counts. Report overall exact location precision with all predicted locations in its denominator, including spurious and wrong-boundary ones. This avoids inventing a slice precision denominator by silently dropping unpaired false positives. For small slices, show raw numerators and denominators. Compare taggers on paired documents and resample documents, rather than individual mentions, if estimating uncertainty.

Finally record whether any word in each gold name splits into multiple subwords. Compare recall and diagnostic counts for split and unsplit names within each seen and unseen slice. Use one common tokenizer for this diagnostic or report each model's fragmentation separately. Recover original word predictions with the declared alignment policy before scoring. Inspect multi-piece errors to separate alignment failures, boundary failures and type failures.

This example checks the slice-assignment pitfall. A full error study additionally needs the document grouping, paired tagger outputs and support counts just described.

The calculations show why the prediction unit matters: the same one-tag error can have different entity-level consequences, and equal conditional error rates can coexist with different precision. Preserve word alignment, sentence identity, mask meanings and metric denominators from preparation through reporting.